

NODES

GE | Euclidean correlator [B] | Euclidean corr. / $G_E(X, X)$ / §5
 GN | Tensor match [B/O] | Conditional tensor-EFT normalisation / $G_N^{\text{nat}} = (8\pi M_G^2)^{-1}$ / after
 physical matching: $G_N^{\text{SI}} = \hbar c^5 / (8\pi (E_{\{M_G\}}[J])^2) / E_{\{M_G\}} \equiv$
 $M_G | \{ \text{joule} \} (\text{GeV} \rightarrow J) / \text{owners Open} \cdot §5.4$
 KG | KG equation [A] | Klein-Gordon / equation / §3.6
 LIV | Cone-speed test [EXT] | Multi-messenger cone test / $|c_{\text{GW}}/c_{\gamma} - 1| \ll 10^{-15}$ / external kill gate ·
 §10
 MBAL | Balance owner [MISSING] | M_{bal} owner / matching bridge [Open] / codimension-3 worldline
 reduction is not a derived spatial string loop / 4D profile,
 stability and common owner are missing · §5.6
 NLO | Orientation stiffness [O] | Candidate orientation stiffness / separate operator / origin Open /
 $\kappa_n \rightarrow M_G^2 \text{ Open} / §5.1$
 P1 | P1: 4D Euclidean [P] | P1 / 4D Euclidean / §2
 P2 | P2: O(4) action [P] | P2 / O(4) action / §2
 P3 | P3: scalar dynamics [P] | P3 / Condensate Φ / §2
 P4 | P4 nonzero chart [P] | P4 / Supplied nonzero-vector / chart / §2
 P5 | P5: medium character [P] | P5 / Medium character / / gauge-completion programme / §2
 P6 | P6: many configs [P] | P6 / Non-uniform / ordering / §2
 PX | Scalar P(X) seed [A/B] | Homogeneous scalar P(X) seed / no independent transverse stiffness /
 A-math conditional · §3
 Ph1 | Clock / EM [O] | $\text{Ph1} / c_{\text{char}} \leftrightarrow c / \text{Open clock} / \text{photon matching} / §4.8$
 Ph2 | Planck match [EXT] | $\text{Ph2} / M_G = M_{\text{Pl}} (\text{obs.}) / §5.4$
 S0 | Action-unit slot [B/O] | Operational action-unit slot S_0 / input to conditional wave / canonical
 / PES formulae / map from either loop candidate Open · §5.6
 S0BAL | Balanced action unit [B/O] | Conditional definition $S_{0,\text{bal}}^{\text{eff}} := \sqrt{(2K_{\theta}(1D)T_{\text{loop}})} /$
 multiwinding is linear / marginal; this is not operational $S_0 /$
 shape, splitting, unwinding, barrier, bending and subtraction
 remain Open · §5.6
 S0EFT | Fixed-core candidate [A/O] | Fixed-core phase-only candidate $S_0^{\text{EFT}} / \text{A mathematics in}$
 supplied 1D core / 4D reduction / nonzero value Open · §5.6
 S0HBAR | Empirical S0 match [B/O] | Explicit empirical calibration $S_0 = \hbar [\text{Level B}] /$
 candidate-to-operational owner map Open / absolute derivation /
 universality Open · §5.6
 SSB | O(3) stabiliser [A/O] | Supplied-vector stabiliser / $O(4) \rightarrow O(3)$ [A-math conditional] / physical
 state / spontaneous selection OPEN · §3
 SU3 | SU(3)c completion [O] | $SU(3)_c [\text{OPEN}] / \dim \text{su}(3) = 8 > 6 = \dim \text{so}(4) / \text{requires extended structure}$
 T1MIN | T1 length minimum [A] | Unique length minimum [A in supplied T1 only] /
 $L_* = \pi |n| \sqrt{(2K_{\theta}/T_{\text{loop}})}$; $S_{n,\text{min}} = 2\pi |n| \sqrt{(2K_{\theta}T_{\text{loop}})} / n=0$ has no finite
 interior minimum · §5.6
 T1MODEL | Supplied T1 loop [B/O] | Supplied T1 one-dimensional loop model [B/Input] / compact θ on
 supplied closed carrier; $K_{\theta}(1D)$, $T_{\text{loop}} > 0 / n \neq 0$, $L > 0$;
 carrier and 4D action owner Open · §5.6
 UC | χ coordinate [A] | Strict-map integrability / one longitudinal χ coordinate / conditional
 kinematics [Level A] · §4.3
 anomaly_arch | Anomaly checks [B] | Anomaly consistency §9.5 / checks only after supplied /
 representations
 boson_hier | Boson inventory [B] | Conditional boson inventory / completion targets only / §Discussion
 common_origin | Common route [O] | Common-medium architecture / physical completions Open /
 §Discussion
 cpt_arch | Discrete symmetries [A/B/O] | Discrete-symmetry programme §9.4 / Euclidean reflections
 exact; / physical P/T/CPT Open
 cpt_open | CPT/anomaly [O] | Discrete-symmetry · Anomaly · Flavour opens / OP-CPT1 · full nonlinear
 CPT / OP-P1 · exact max parity / OP-BRANCH1 · branch-pair cosmology /
 OP-GUT6c · mixed U(1)-grav (emergent) / OP-Yukawa · CKM/PMNS/CP phases/ ν
 spectrum
 cstar | Scalar cone slope [A/O] | $\hat{c}^* = \sqrt{(\beta/(\alpha-\beta))}$ / supplied-operator slope [Level A conditional] /
 physical c_{char} / cross-sector matching Open · §5.3
 dirac | Spinor / Dirac [B/O] | Supplied spinor / Dirac / conditional standard form / physical fermions
 Open · §9
 einstein | Einstein target [B] | Linear Einstein target / supplied tensor completion / §6
 fifth | Director operator [B/O] | Allowed director operator / $\mu_5 \bar{\psi} \gamma^5 \text{An}_{\Lambda} \psi$ / coefficient / observable
 Open · §7.6
 flavour_arch | Flavour programme [O] | Flavour programme §9.7 / overlap / seesaw ansätze supplied; /
 CKM / PMNS hierarchy Open
 gauge | Gauge completion [A/B/O] | Gauge-completion programme / global phase algebra exact; / local
 bundles / photon / charges Open / §7
 goldstone | One-mode EFT [B/O] | Physical one-mode EFT [Level B/Open] / requires supplied reduced
 action / domain / projector / §4
 lorentz | Scalar operator [A/O] | Supplied scalar principal operator / one cone [Level A conditional]
 / physical c_{char} / cross-sector cones Open · §3.6
 neutrino | Neutrino programme [O] | Neutrino model-building §8.8 / seesaw operator / states supplied /

geometric combinations are anchors only / masses / PMNS / Majorana
status Open
parity | Vector stabiliser [A/O] | Vector stabiliser = SU(2)_diag / selects neither chiral factor /
physical weak bundle / chirality Open · §9
schrodinger | Schrodinger envelope [B/O] | Schrödinger-type envelope / conditional state / operational
S_0 map / physical owner Open · QS
second_law | Second-law target [O] | ECT thermodynamic / second-law theorem Open / §3.9
skyrmion | Texture sectors [A/O] | Texture sectors / $\pi_3(S^3) = \mathbb{Z}$ labels / two-derivative $E_2 \propto R$
collapses / stabilising action/mass Open / §8
threegen | Three generations [O] | 3 generations / [OPEN]
topo_anti | Supplied S3 target [A/O] | Supplied S^3 target [A-math conditional] / $\pi_0 = \pi_1 = \pi_2 = 0$ /
topological classes only; physical defects / relic absence OPEN
/ secondary sectors separate · §8
unruh | Unruh/KMS target [A/O] | Standard Unruh/KMS target / A external; ECT state / detector Open /
QS

VISIBLE DIRECTED EDGES

E001 | GN -> LIV [dashed] | (no literal edge label)
E002 | GN -> einstein [dashed] | tensor completion missing
E003 | KG -> schrodinger [dashed] | NR state + S0 match
E004 | MBAL -> S0 [dashed] | owner map Open
E005 | NLO -> GN [dashed] | (no literal edge label)
E006 | P1 -> SSB [dashed] | (no literal edge label)
E007 | P2 -> SSB [dashed] | (no literal edge label)
E008 | P3 -> GE [dashed] | (no literal edge label)
E009 | P3 -> SSB [dashed] | (no literal edge label)
E010 | P4 -> SSB [solid] | (no literal edge label)
E011 | P4 -> lorentz [dashed] | (no literal edge label)
E012 | P5 -> gauge [dashed] | (no literal edge label)
E013 | P6 -> SSB [dashed] | (no literal edge label)
E014 | P6 -> skyrmion [dashed] | (no literal edge label)
E015 | P6 -> topo_anti [dashed] | (no literal edge label)
E016 | PX -> NLO [dashed] | separate operator; no P(X) derivation
E017 | Ph1 -> cstar [dashed] | (no literal edge label)
E018 | Ph2 -> GN [dashed] | (no literal edge label)
E019 | S0 -> KG [dashed] | operational action unit
E020 | S0 -> S0HBAR [dashed] | empirical calibration [Level B]
E021 | S0BAL -> MBAL [dashed] | owner not derived
E022 | S0EFT -> S0 [dashed] | owner map Open
E023 | SSB -> GE [dashed] | (no literal edge label)
E024 | SSB -> UC [solid] | (no literal edge label)
E025 | SSB -> skyrmion [solid] | (no literal edge label)
E026 | SSB -> topo_anti [solid] | (no literal edge label)
E027 | T1MIN -> S0BAL [solid] | (no literal edge label)
E028 | T1MODEL -> T1MIN [solid] | (no literal edge label)
E029 | UC -> PX [solid] | (no literal edge label)
E030 | UC -> S0EFT [dashed] | supplied 1D fixed-core reduction
E031 | UC -> T1MODEL [dashed] | supplied conditional model
E032 | UC -> cstar [dashed] | separate supplied operator
E033 | UC -> goldstone [dashed] | requires action / domain / projector
E034 | UC -> lorentz [dashed] | separate supplied operator
E035 | anomaly_arch -> cpt_open [dashed] | (no literal edge label)
E036 | anomaly_arch -> flavour_arch [dashed] | (no literal edge label)
E037 | cpt_arch -> anomaly_arch [dashed] | (no literal edge label)
E038 | cpt_arch -> cpt_open [dashed] | (no literal edge label)
E039 | cstar -> GN [dashed] | (no literal edge label)
E040 | cstar -> LIV [dashed] | (no literal edge label)
E041 | dirac -> fifth [dashed] | (no literal edge label)
E042 | dirac -> flavour_arch [dashed] | (no literal edge label)
E043 | dirac -> neutrino [dashed] | (no literal edge label)
E044 | einstein -> second_law [dashed] | (no literal edge label)
E045 | fifth -> common_origin [dashed] | (no literal edge label)
E046 | fifth -> cpt_arch [dashed] | (no literal edge label)
E047 | flavour_arch -> cpt_open [dashed] | (no literal edge label)
E048 | gauge -> SU3 [dashed] | (no literal edge label)
E049 | gauge -> anomaly_arch [dashed] | (no literal edge label)
E050 | gauge -> boson_hier [dashed] | (no literal edge label)
E051 | gauge -> common_origin [dashed] | (no literal edge label)
E052 | gauge -> cpt_arch [dashed] | (no literal edge label)
E053 | gauge -> dirac [dashed] | (no literal edge label)
E054 | gauge -> neutrino [dashed] | (no literal edge label)
E055 | gauge -> parity [dashed] | (no literal edge label)
E056 | gauge -> threegen [dashed] | (no literal edge label)
E057 | goldstone -> boson_hier [dashed] | (no literal edge label)
E058 | lorentz -> KG [solid] | (no literal edge label)
E059 | lorentz -> einstein [dashed] | (no literal edge label)

E060 | lorentz -> gauge [dashed] | (no literal edge label)
E061 | lorentz -> unruh [dashed] | (no literal edge label)
E062 | parity -> common_origin [dashed] | (no literal edge label)
E063 | parity -> cpt_arch [dashed] | (no literal edge label)